

Instruction Manual

Stord Twin Screw Press

Type MS-64SF

Gear Ratio 1 : 125

January, 1994

Please note:

When ordering spare parts
please quote all
specifications and
Machine No.:

Stord International A/S
Sandslimarka 51
P. O. Box 9
N-5049 Sandsli - Norway
Teleph.: +47 55 22 36 00
Telefax: +47 55 22 36 01
Telex: 42051

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Type MS-64SF
Instruction Manual**

Contents: Page:

<u>1. Description</u>	4
. Working Principle	4
. Cover Guards	4
. Press Cage	5
. Inlet Chute, Outlet Chute	5
. Press Screws	5
. Bearings and Stay Bars	5
. Driving Gear	6
. Frame	6
 <u>2. Installation of the press (Instalación de la prensa)</u>	 7
. Alignment of the press	10
. Lubrication	11
. Lubrication of Gearbox	11
. Lubrication Control and Protection	12
. Starting Up at Low Temperatures	12
. Press screw Bearings, Inlet End	12
. Oil seals on Output Shafts from Gearbox	12
 <u>3. Operation</u>	 13
 <u>4. Inspection and maintenance</u>	 14
. 1. During operation	14
. 2. After production	14
. Cleaning of the press cage	14
. Strainer Plates	15
. Painting	15
. Press Screw Flights	15
. Lubrication of Gearbox	16
. Protection of the Gearbox in Lay-off Periods	16
 <u>5. Dismantling and remounting</u>	 17
. Dismantling of the Cover Guards	17
. Dismantling and Remounting of Upper Part of the Press Cage	17
. Removal of Inlet Chute	17
. Removal and Remounting of Backing Plates	18
. Removal of Strainer Plates	18

. Replacement of Strainer Plates	18
. Spot-welding of Strainer Plates	19
. Riveting of Strainer Plates	19
. Removal of Press Screws	20
. Remounting of Press Screws	21
. Replacement of Oil Seals on Gearbox Output Shafts	21
. Dismantling of Gearbox	22
. Assembly of Gearbox	23
. Fitting of Roller Bearings	24
 6. Removal and fitting of shafts in the lower section of the gearbox	
. General	26
. Removal of input shaft	26
. Removal of shaft No. 2	26
. Refitting of shaft No. 2	27
. Repositioning of input shaft	27
 7. Data sheet	28
. Weight of parts	29
. List of parts – press	30
. List of spare parts (recommended to be kept in stock)	34
. List of tools	35
 8. Drawings	
. General arrangement	36
. Main Drawing	37
. Section of gearbox	38
. Cross section & Details	39
. Replacement of Strainer Plates	40
. Repair of Strainer Plates	41
. Riveting Knob and Drifts	42
. Packing rings on Gearbox Output Shafts	43
. Dismant. & Repl. of 2nd Shaft & Wheel	44
. Rebuilding of Worn Screw Flight Edges	45
. Flow diagram of lubricating oil	46
. Lubrication Chart	47
. Lubricating oil pump	48
. Lubricating oil filter	49

1. Description

2. Installation of the press

3. Operation

4. Inspection and maintenance

5. Dismantling and remounting

**6. Removal and fitting of shafts
in the lower section of the gearbox**

7. Data sheet, List of parts

8. Drawings

9. Driving arrangement (if delivered)

10.

Stord Twin Screw Press

Type MS-64SF

Description

Working Principle:

The wet material to be pressed is fed into the inlet chute at the end of the press, where it reaches the active pressing device consisting of two slowly rotating screws. The wet material fills the free space between the screw flights, and the dewatering is achieved in the following way. By rotation of the screws the material is conveyed into the closely fitted strainer housing and simultaneously submitted to a steadily increasing pressure, which causes the liquid to drain off through the perforated strainer plates.

The pressure is produced by the screws, which are designed so that the channel, i.e. the space between the flights, through which the material is forced steadily narrows from intake to outlet end. When the pressing process is concluded the material leaves the press through the outlet chute.

The strainer housing, i.e. the press cage, comprises the internal strainer plates, the backing plates and the relevant clamping structure. Underneath the cage there is a tray collecting the press liquid, allowing it to drain through the flanged outlet at the bottom.

Due to the intermeshing of the flights of the two counterrotating screws, the material is prevented from sticking to the screws and rotating with them. Axial progress of the material is secured, and no clogging occurs.

The rotation of the screws, which are placed parallel to each other, is effected by a motor connected to the gearbox. The power is divided in the first stage of gears and transmitted via two further stages to each of the press screws.

Cover Guards:

The press cage is covered by hatches, both on the top and each side of the press. The top hatches are fixed by staybars in upper cages, and running along the length of the press. Each hatch has an overlapping edge to the next.

The side hatches also overlap each other, and are fixed to the frame and top hatches by guide screws and cover locks. Each side hatch at the outlet end has inspection shutter.

Press Cage:

For easy dismantling and access to the press screws, the cage is split longitudinally and consists of an "upper part" which can be lifted off, and a "lower part" fixed to the frame. The cage parts are clamped together by means of forked sling clamps. See drwg. No. 4I-35247. The drawing shows that the cage is composed of the following parts. Around the press screw are fine perforated strainer plates which are fastened to the heavy external backing plates. These are kept in position by strongly built "bridges" arranged across the plates, together with longitudinal "side bars" and "center bars". To ensure rigidity, staybolts and spacing pipes are inserted between the bridges.

The upper part of the cage is accurately positioned on the lower part by the side bars, which are guided by the lower bridges.

Inlet Chute, Outlet Chute:

The inlet chute has perforated side walls and is fastened with screws to the side bars of the lower cage. A window in the wall permits inspection of the material flow and the charge of the press.

The outlet chute is attached to the end bridge, the gearbox and both sides of the press frame.

Press Screws:

The two press screws are fabricated hollow trunks and have helical flights along. During operation the screws rotate in opposite directions and have therefore right hand and left hand flights respectively. At the inlet end the trunk extends into the shaft journal of the inlet end bearing. At the other end (outlet end) it is provided with a machined flange for screw connection with the driving shaft.

Bearings and Stay Bars:

Bearings in the gearbox are of the spherical roller type partly splash lubricated, partly lubricated by a full flow oil system. The press screw bearings at the inlet end are also spherical roller bearings type and lubricated by grease.

To cope with the great axial forces from the twin screws, heavy spherical roller thrust bearings are used. These are fully flow lubricated from the oil feed system and are spring loaded for idle operation. Between the gearbox and the press cage are two solid staybars mounted to counteract the axial forces.

The press must never be put in operation without these staybars fitted.

For further particulars see instructions for dismantling and reassembly.

Driving Gear:

For speed reduction a totally enclosed gearbox is arranged at the end of the press frame. The overall reduction ratio of the gearbox is 125 : 1. Pinions and spur gears as well as roller bearings are automatically oil lubricated.

The press is arranged for V-belt drive (or chain drive) from an electric motor with variable speed, or from a fixed speed motor in connection with a suitable variator, if needed. Estimated power consumption is specified on the Data Sheet.

For further particulars see instruction for dismantling and reassembly.

Frame:

The press frame, which carries all the vital parts of the press, is made of a simple and rigid mild steel construction supported by four legs and the gearbox. The frame is bolted onto the gearbox by a special supporting bracket.

Instalación de la prensa

Es indispensable que el fundamento sobre el que se basarán la prensa y su motor tenga una construcción pesada y rígida, y que sea construido de acuerdo con el plano de dimensiones de Stord International.

Para facilitar el transporte, la prensa podrá desmantelarse parcialmente antes de salir del taller, donde ha sido sometida a un período de funcionamiento de prueba antes de ser inspeccionada, desmontada y embalada para el transporte.

Al instalar más de una prensa a la vez, tener cuidado de no mezclar componentes que pertenecen a prensas con distintos números de producción.

Para facilitar el manejo y levantamiento de las piezas individuales, es necesario instalar una grúa o aparejo por encima de la prensa.

La capacidad mínima de carga recomendada es de 7 toneladas.

Comprobar que todas las superficies mecanizadas están limpias y sin daños.

El montaje y adaptación de la prensa en el sitio se lleva a cabo de la forma siguiente:

1. Comprobar que el fundamento prefabricado está en la posición horizontal justa.

Retirar las cubiertas y las secciones superiores de la carcasa.

Situar la prensa sobre el fundamento prefabricado. Si éste es de hormigón con agujeros para los pernos, se vertirá cemento blando apropiado en los agujeros bajo la placa de base. Entonces el montaje sólo podrá continuarse después de un período de 3 días de secado.

2. La prensa debe ser exactamente nivelada mediante instrumentos ópticos, véase plano número 21-41829.

El punto de medición para prensas nuevas es el bastidor de la prensa, y para prensas de segunda mano, las barras laterales.

Nivelar un lado primero, y luego el lado opuesto, usando los tornillos de ajuste de la caja del reductor de velocidad y soportes. Cuando el lado opuesto esté correcto, vuelva a comprobar el primer lado para ver si es necesario hacer nuevos ajustes.

Tres puntos de medición en cada lado de la prensa serán suficientes.

3. Poner separadores de acero por debajo del reductor de velocidad y el soporte. Los separadores de acero deben estar justos y cubrir el área de la placa de base.

Se aprietan los pernos de fundamentación.

Volver a comprobar la horizontalidad, realizando cualquier ajuste que resulte necesario.

Los apartados 4 y 5 siguientes deben seguirse cuando los tornillos de la prensa han sido desmontados o enviados por separado.

4. Situar el tornillo de la prensa en la carcasa tal como se indica en el dibujo, produciendo que las hélices de los tornillos estén correctamente engranadas, o sea, que cada hélice o filete de un tornillo debe estar a la mitad de la distancia entre dos hélices del otro tornillo.

Ahora se puede empujar cada tornillo sobre el eje hasta que toquen la brida, en la que se fijan cuidadosamente los pernos roscados. Asegúrese de que las caras contiguas de las bridas están completamente limpias, y que las marcas del cuerpo de los pernos corresponden con las marcas de la brida.

Es muy importante que los pernos de la brida M36 sean apretados a 2.200 Nm (1.600 libras-pie). En caso de que no resulte posible comprobar el grado de ajuste, el ángulo de incidencia ("vuelta de la tuerca"), podrá medirse desde la instalación fija de los pernos (ajusto fino). Para esta dimension de pernos en particular, de $\frac{1}{2}$ a $\frac{3}{4}$ vuelta. Los pernos se aseguran con placas de bloqueo.

5. Los tornillos de la prensa se levantan por el lado de entrada, y la sección inferior de las cajas de cojinete se monta en la prensa mediante las clavijas y los pernos. El manguito de empuje y los cojinetes se montan en el eje, y cada tornillo se bajan a su posición correcta.

Montar los cojinetes de la mitad de la carcasa y fijarlos mediante los manguitos adaptadores. Montar la sección superior de la carcasa, y llenar con grasa. Luego montar la tolva de entrada. La placa del extremo se fija mediante clavijas cónicas. Después se montan los anillos de sello y la caja de sello en cada gorrón del eje.

6. Limpiar con aire comprimido la sección inferior de la carcasa de la prensa para eliminar suciedades e impurezas.
7. Hacer girar los tornillos de la prensa y comprobar mediante un calibre de láminas el juego entre las hélices y las placas perforadas de la sección inferior de la carcasa en toda la longitud de los tornillos. El juego debe ser como mínimo 0,8 mm (0.03").

Prestar atención especial al juego en la sección media de la parte inferior de la carcasa. Podrá ser necesario bajar la parte media de la prensa para conseguir un juego satisfactorio en este lugar. Ahora pueden apretarse los pernos de fundamentación de los pies de la sección media.

8. Posicionar las 2 secciones superiores de la carcasa de la prensa. Introducir las barras de separación y fijarlos. Fijar la sección superior de la carcasa de la prensa mediante los sujetadores. Estos deben apretarse bien.
9. Para facilitar el desmontaje futuro de la tolva de entrada de la prensa, debe quedar abundante espacio libre (al menos 470 mm) por encima de la brida de la entrada de la prensa al desmontar la tolva exterior.
10. Hacer girar manualmente el eje de entrada del reductor de velocidad. Este debe girar fácil y regularmente. El juego entre los tornillos y la carcasa superior se comprueba visualmente con una lámpara de mano, mirando a través de la placa perforada y girando la prensa lentamente a mano. Escuchar si hay sonidos que indiquen que el tornillo toca la placa perforada. El juego también puede comprobarse mediante un indicador de profundidad especial, insertándolo a través de los orificios.
11. Comprobar que todos los pernos y tuercas están bien apretados.
12. Montar el motor de impulsión y el motor de la bomba de aceite. Los arrancadores eléctricos de la unidad hidráulica de la prensa y la bomba de lubricación deben estar conectados de modo que la unidad hidráulica y el motor de impulsión sólo puedan arrancar cuando ya está funcionando la bomba de lubricación. El motor de impulsión debe pararse automáticamente se se detiene el motor de la bomba de lubricación.
13. Rellenar con aceite de acuerdo con las instrucciones de lubricación. Para la unidad hidráulica, ver instrucciones separadas.

¡OJO! Comprobar que las tuberías de aceite de la unidad hidráulica están completamente limpias.
14. Montar las cubiertas de la carcasa de la prensa. Para prensas provistas de tuberías para lavado con agua, éstas deben montarse antes de que se pongan las cubiertas.
15. Hacer funcionar la prensa a modo de prueba acoplando el motor, y desacoplar inmediatamente si algo parece mal. Inspeccionar y volver a poner en marcha la prensa.

Al hacer funcionar la prensa en temperaturas ambientales de menos 10 °C (50 °F), se recomienda omitir el filtro durante unas 12 horas, con el fin de evitar que el aceite frío y espeso cause daños en el cartucho.

Installation of the press

It is essential that the foundation on which the press and its motor are based, is of a heavy and rigid construction and is made according to the dimension drawing from Stord International.

To facilitate the transport, the press can be partly dismantled before dispatch from the workshop, where it has been run in and checked before being dismantled and packed for shipment.

When fitting up more than one press simultaneously, one should take care not to mix components from presses of different production numbers.

To facilitate handling and lifting of the separate parts, a travelling hoist or tackle must be installed above the press.

Recommended lifting capacity is minimum 7 tons.

Check that all machined surfaces are clean and without damages.

Mounting and fitting of the press on site are carried out as follows:

1. Check that the premade foundation is in level.

Take off the covers and the upper cages.

Place the press on the premade foundation. If this foundation is made of concrete with holes for the bolts, suitable soft cement must be poured into the holes under the baseplate. The assembly can be continued only after a drying period of min. 3 days.

2. The press should be accurately levelled by means of optic levelling instrument, please see drawing no. 21-41829.

The measuring point for new presses is the press frame, and for secondhand presses the side bars.

Level one side first, and then the opposite side by using the adjustment screws on gearbox and supports. When the opposite side is correct, please check the starting side for any adjustments.

Three measuring points on each side of the press is satisfactory.

3. Steel spacers should be put under the gearbox and support. The steel spacers have to be accurate and cover the area of the bedplate.

The foundation bolts to be tightened.

Please check the levelling again and make adjustments if necessary.

The following sections 4 and 5 should be followed when the press screws are dismantled or sent separately.

4. Place the press screw in the cage as indicated on the drawing, and take care that the screw flights mesh correctly, i.e. each flight of the one screw must be in position midway between two flights of the other screw.

Each press screw can now be pushed onto the shaft journal until it lies true against the flange, whereupon the screw bolts are fastened thoroughly. Make sure that the abutting flange faces are absolutely clean and that the markings on the screw body correspond with those on the flange.

It is very important that the flange screws M36 are torqued to 2.200 Nm (1.600 ft. lbf.). Should it not be possible to check the degree of adjustment, the angle of incidence ("turn of nut") can be measured from the fixed installation of the screws ("snug fit"). For this particular dimension of screws, $\frac{1}{2}$ – $\frac{3}{4}$ turns. The screws are secured by lock plates.

5. The press screws are lifted at the inlet end, and the lower part of the bearing housings is fitted onto the press by means of dowels and bolts. The thrust sleeve and the bearings are fitted onto the shafts, and each screw is lowered into place.

Fit the bearings in the middle of the housing and fix by means of the adapter sleeves. Fit the upper part of the housing and fill with grease, then fit the inlet chute. The end plate is fixed by means of conical dowels. Then fit the seal rings and seal house onto each shaft journal.

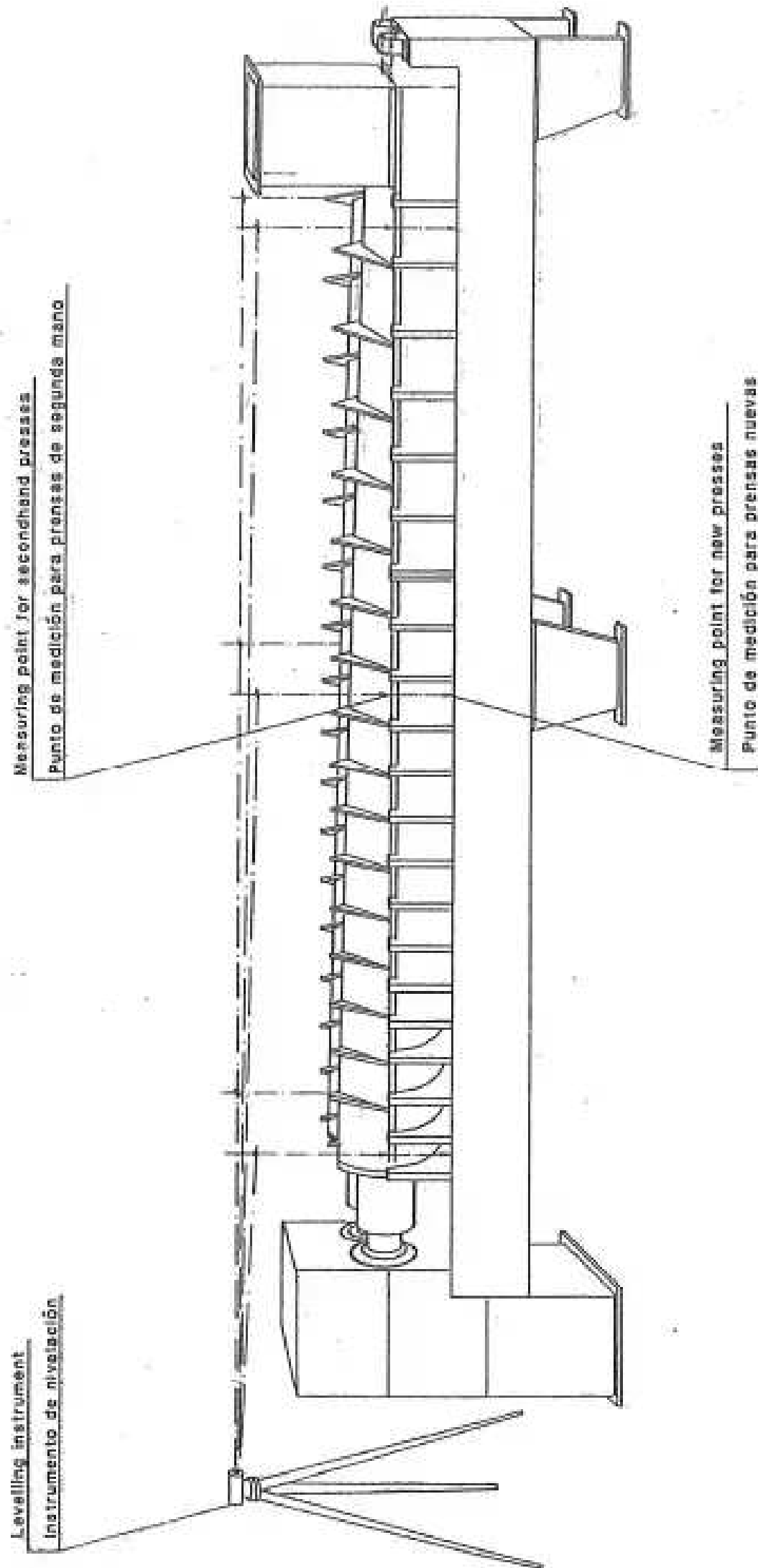
6. Air blast the lower part of the press cage to remove dirt and foreign matters.
7. Turn the press screws and check by means of a feeler the clearance between the flights and the strainer plates in the lower part of the cage all over the length of the screws. The clearance should be minimum 0,8 mm (0,03").

Pay special attention to the clearances in the lower cage in the mid section. It may be necessary to lower the press on the middle to obtain satisfactory clearance here. Foundation bolts on the middle legs can now be tightened.

8. Position the 2 upper parts of the press cage. Insert the distance bars and fasten them. Clamp the upper part of the press cage to the lower part by means of the sling clamps. These should be pulled tight.
9. To facilitate future dismantling of inlet chute in press, there should be arranged for ample free height (at least 470 mm) above the flanged inlet on the press, when the outside chute is removed.
10. Turn the gearbox input shaft by hand. This should turn easy and even. The clearance between screws and upper cage is checked visually with a hand lamp by looking through the perforated plate and turning the press slowly by hand. Listen for sounds indicating the screw touching the strainer plate. The clearance can also be checked by means of a special depth indicator inserted through the perforations.
11. Make sure that all bolts and nuts are tightened.
12. Fit the driving motor and motor for oil pump. The electrical starters for hydraulic unit for press and oil lubrication pump must be connected in a way that the hydraulic unit/driving motor should only be able to start after the oil lubrication pump is running. The driving motor should automatically be switched off if the oil lubrication pump motor cuts out.
13. Fill oil according to lubrication instructions. For the hydraulic drive unit, please see separate instructions.

NB! Please make sure that oil pipes for hydraulic drive are completely cleaned.
14. Mount the covers over the press cage. For presses equipped with flushing pipes, these should be fitted before the covers are positioned.
15. Start the press for trial running by engaging the motor, and cut out immediately if anything appears to be wrong. Check and start again.

When starting press in ambient temperatures below 10 °C (50 °F), it is recommended to by-pass the filter for about 1-2 hours, in order to prevent damage to the cartridge by the thick cold oil.

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Lubrication

Lubrication of Gearbox:

The gearbox has been thoroughly cleaned before dispatch from the works and is ready for filling up with the required quality and quantity of oil up to the middle of the oil level indicator.

The gearbox, bearings, gearwheels and lubrication system have been treated with a rust-inhibiting oil which does not require cleaning off before filling the oil. The recommended quantity and quality is to be found on a separate Lubrication Chart further back in this instruction manual. The gearbox has its own lubricant system. The oil pump sucks from the oil pan in the gearbox and pumps it through the strainer to an oil distributing tank in the upper part of the gearbox.

From the oil tank the oil is distributed over the gear wheels and to the bearings and flows back to the oil pan.

Regular checks on the oil pressure from the pump and flow of oil from the two large thrust bearings are recommended. (Visual inspection through the transparent side covers).

For filtration of the oil, the system is equipped with filters. One coarse (inside the gearbox) on the suction side of the pump and a finer one on the delivery side. The finer filter has a valve allowing the oil to flow in by-pass during cleaning of the filter and bowl.

During running-in of the press it is recommended to clean the external filter after the first 30 hours. Later every 200 hours, or when the gauge shows pressure in excess of 2,8 bar (40 psi).

(When starting up with cold oil the pressure will initially exceed this value). Cleaning is done by loosening down the bowl and cleaning the brass gauze cartridge in petrol or paraffin. For cleaning use a soft brush. Wipe off the magnet with a clean rag, check that the filter cartridge is in working order, replace the filter and reset valve for filtration. Spare cartridges should always be in store.

The oil in the gearbox should be changed for the first time after 2000 working hours, later after every 5000 hours or according to sample analysis by the oil company. The internal suction strainer should be cleaned at every oil change.

If the oil temperature exceeds 60 °C (140 °F), it will be necessary to install an oil-cooler. Particulars from Stord International A/S.

The oil pump motor is to be lubricated according to normal practice.

Lubrication Control and Protection:

If the press is to be run without regular supervision, it is recommended to fit a pressure switch with a warning and motor cut out system.

The external oil pipe is in front of the oil filter equipped with a 3/4" int. pipe thread junction for the possible fitting of a pressure switch. On delivery, a 3/4" blanking plug is screwed in. The warning system should give alarm at oil pressure below 0,25 bar (3,5 psi) and the interloc. should automatically shut down the driving motor.

Pressures above 2,8 bar (40 psi) indicates that the oil filter is due for cleaning.

(When starting up with cold oil, the pressure will initially exceed the above mentioned value.)

Starting up at low Temperatures:

At low ambient temperatures, 5 °C (40 °F) or lower, the oil can be so viscous that the pump will have difficulties in pumping the lubricant and in this case the oil should be heated some-what, primarily at the pump intake. This can be done for instance by using a large fan heater to blow hot air against the gearbox at the pump intake. If the press is erected outside, a preheater or an electrical spiralheater should be installed.

Press Screw Bearings Inlet End:

The two bearings at the inlet end are lubricated by grease. Fill the housing completely with grease when fitting this and refill once every other week with a grease gun. The bearings should be cleaned and replenished with grease annually.

Oilseals on Output Shafts from Gearbox:

It is recommended to grease these seals once a year.

This is done by unscrewing both of the plugs in the cover (item 14) and fitting a standard grease nipple in one of the holes.

Press grease until it flows out of relief hole. Now refit the plugs.

Operation

When the erection of the machine is completed and oil filled, start the press and let it run for about 1 hour before charging. When starting press in ambient temperatures below 10 °C it is recommended to by-pass the filter for about 1-2 hours. This to prevent damage to the cartridge by the thick cold oil. Start feeding with the motor running at lowest speed, and be sure that there is a suitable level of raw material in the inlet chute. Generally normal pressing conditions are not obtained before the press is completely filled and the initial charge has gone through. If feeding is insufficient, the pressure will decrease and the water content in the discharge will increase. The best results are obtained when the press is working at a constant speed with a continuous and uniform supply.

Watch the kW or Amp. meter and make sure that the electrical loading is not higher than the electric motor rate.

To avoid risk of damaging the machine as a result of clogging and encrusting, the press should be emptied when taken out of operation. However, if the press is stopped for emergency not exceeding half an hour, the material may be left in the press.

If the press stops due to overload/jamming, lift up the upper part of the press cage (at the outlet end), allowing some of the material to be removed before the press is put into operation again.

IMPORTANT!

Do not in any case try to run the press screws in reverse, as severe damage of the gearbox then can occur.

Generally, stones and small pieces of iron will do no harm to the press. Bigger pieces, however, might cause damage to the strainer plates. In case of damage, stop the press at once, lift off the top part of the press cage, and remove the material. Replace strainer plates as described in special instruction. In case of minor damage to the strainer screen the press can be run empty, and by provisionally blinding off the damaged area from the outside, the press can quickly be put in operation again.

Inspection and maintenance

1. During Operation:

From time to time the press should be inspected and fasteners tightened if necessary. Deposits of material on the press cage, frame or other places should be hosed off.

The oil pressure and temperature should be checked regularly. The filter should be cleaned for the first time after 30 operating hours. Later – under normal operating conditions – every 200 hrs. Should the oil pressure exceed 2,8 bar (40 psi) the filter should in any case be cleaned or changed.

The oil in the gearbox should be changed for the first time after, say, 2000 operation hrs. Later every 5000 hrs or according to analysis.

2. After Production:

For extended shut-down periods, the press should be hosed with water [pressure up to 50 bar (750 psi)] in order to avoid material remainder fermenting and causing corrosion. Further cleaning and maintenance should take place as soon as possible.

Cover guard sections and shutters should be properly cleaned and brushed on the inside. Hot water should be used. Check that the seals are in order.

Once a year the press cage should be opened for examination of the strainer plates.

A suitable, spacious place in the works should be reserved for the placing of parts during overhaul work. Special attention should be paid to machined parts, which should be placed on wood supports.

Cleaning of the Press Cage:

The upper part of the cage sections should be detached and cleaned without further dismantling unless damaged strainer plates have to be replaced or if there should be signs indicating corrosion on the contact surface between the backing plates and the bridges. Hot water should be used for cleaning, preferably at high pressure. Material remainder sticking to the perforation can be removed by means of a steel brush and a scraper and blown out by pressurized air or water.

The easiest way to clean the lower part of the cage is to remove the backing plates. See section: Removal and Remounting of Backing plates.

Frames with strainer plates under the inlet chute can be cleaned in position in the press. However, these frames can easily be removed after the inlet chute has been dismantled. Before assembly, it is important that the surfaces of the bridges mating the bars and grooves in the side bars are thoroughly cleaned, and care must be taken to ensure that the plates are mounted in the correct place.

Strainer Plates:

Minor damage to the strainer plates can often be made good by welding (e.g. Argonarc method). Afterwards the welding is ground down carefully.

Painting:

Good maintenance of the press is worthwhile, and too much time should not elapse before it is repainted. It is important that grease, dirt and rust are removed first. It is advisable to use a good primer. Information on suitable qualities of paint can be supplied upon request.

Press Screw Flights:

The top of the press screw flights are on some materials subject to wear and the leading edge should be rebuilt by welding when so worn that the pressing effect is influenced. This depends somewhat on the nature and condition of the material to be pressed.

Before welding takes place, a suitable backing strip should be clamped to the screw flight, as shown on fig. 41-33430. This strip may be of copper (not annealed) or carbon. The latter is preferable, because the weld deposit sometimes has a tendency to stick to the copper backing. This may, however, be avoided to some degree if the arc is not directed at the copper. As regards the choice of welding electrodes, we are always at your service.

Lubrication of Gearbox:

The gearbox should be flushed thoroughly when the oil is changed. Clean cloths of cotton or linen can be used for drying. Cotton waste (twist) must on no account be used.

The type of oil is specified in the lubrication chart.

Protection of the Gearbox in Lay-off Periods:

Due to condensation and possible acid formation internally in the gearbox during lay-off periods, it is necessary to protect roller bearings, gearwheels and lubrication system against corrosion. Preparing for a long lay-off period, the oil should therefore be drained immediately after the press has been taken out of commission, and the gearbox and filters should be cleaned. The gearbox is then filled with rust-inhibiting oil (for instance Mobilarma 524) to the normal oil level. The press is then run idle for 10–15 minutes with this oil, the oil is then drained and can be used in the next press or taken care of till next time. It is not necessary to clean off this oil before normal lubricant is filled, before putting the press into commission again.

Alternatively, the gearbox can be protected against rusting by leaving the lubrication oil in and then once every two weeks, starting the press and idling it for 5–10 minutes. Before starting, drain out possible water that could have gathered in the gearbox.

Dismantling and remounting

Dismantling of the Cover Guard:

Remove the side hatches first and then the top section.

Clean the covers.

Rinse the press cage before stripping. (This should always be done after the press has been put out of commission to prevent clogging of the holes in the cage plates).

Dismantling and Remounting of Upper Part of the Press Cage:

The press cage consists of an upper part and a lower part arranged as mentioned in the above description and shown on drwg. No. 41-35247. Remove the upper stay bar at the discharge outlet.

The upper part of the cage is split vertically in the middle. To remove the two parts of the cage, swing down the forked sling clamps and loosen the screws from the split bridge. The "Outlet End Part" can now be lifted off.

If it is desired to lift off the inlet end part of the cage, the other part must be loosened and moved away from the inlet chute before lifting off. Alternatively the inlet chute may be taken off, which also gives the necessary space.

When remounting the press cage, check that the bridges in the upper and lower part correspond accurately. The stay bar should be tightened so that it is in compression and no clearance is to be found. See that the surfaces of the side bars are clean, tighten well all the clamping screws, and ensure that there is no clearance between the upper and lower side bars.

Removal of Inlet Chute:

When the press inlet chute is to be removed, some free space above the press chute is needed for lifting. (Minimum 470 mm - 19 inches). Disconnect top flange of the chute, and remove the lower flange screws away from the side bars and the end plate. Lift the chute carefully away. When remounting, clean adjoining surfaces and see that the chute fits well and is clear of press screws before tightening the fixing screws.

Removal and Remounting of Backing Plates:

In order to take out the plates from the cage it is necessary first to unscrew the side bars from the bridges to the plates in question (see drwg. 41-35247). The plates are then loose and can be removed. It is not necessary to loosen the center bar.

We advise to mark each plate when removing, in order to ensure correct refit.

The plates in the lower part of the cage may be removed without removing the press screws. A suitable crowbar may be useful to get hold of the backing plates and raise them, as indicated by dotted lines in drwg. 41-35247. Be careful that the point of the crowbar does not damage the strainer plate. Take care also of the arched distance pieces between the backing plates.

When a frame under the feeding chute is to be taken out, the chute has to be removed and the upper cage, if not already dismantled, moved along sufficiently to free the side bars.

The outer surfaces of the backing plates are machined accurately to the diameter of the corresponding bridges, and it is consequently of great importance that they are placed in the original position when reassembling. Check by means of a gauge that there is no clearance between backing plates and bridges.

When fixing the side bars, make sure that the screws are pulled tight. Some gentle blows with a soft mallet near the screws at the same time as the screws are tightened, may be effective.

Removal of Strainer Plates:

The strainer plates are spot-welded or riveted to backing plates. They are usually spot-welded to backing plates of stainless steel, drawing No. 41-34405, and riveted to mild steel backing plates, drawing No. 41-33432.

When replacing a strainer plate spot welds are broken with a flat chisel, rivets are drilled out.

When the strainer plate is removed, the backing plate ought to be cleaned with a wire brush and washed. Remove remaining spots by careful grinding.

Replacement of Strainer Plates:

Spare strainer plates should always be kept in stock at the factory. On request, however, such plates are supplied at short notice, cut to accurate size and perforated as prescribed, but not rolled.

The strainer plates are spot-welded or riveted to backing plates.

Spot-welding (for stainless backing plates only):

The strainer plate should be preformed by rolling and clamped to the backing plate in a simple fixture, as shown on drwg. 41-34405, after which spot-welding takes place.

The clamping fixture consists of two semi-circular pieces of wood with the same radius as the press screw + 1 millimetre, 2 tension girths with threads and nuts, and 2 cross bars. The figure shows the assembly of the plates in the fixture.

Before welding check that the strainer plate is in the correct position on the backing plate.

The first spot weld should be located nearest to the middle of the arched edges of the plates – in both ends, proceed towards the edges.

When the welding has been done, bend the protruding edges of the strainer plate over the backing plate. At the arched ends the strainer plate should be split between the welds to facilitate bending – see drawing. At each of the four corners a small superfluous piece of the plate has to be cut away. The edges should be hammered thoroughly.

Finally the surplus material of the strainer plate edges must be well trimmed.

Replacement of strainer plates on the frame positioned under the feeding chute, is carried out in the same way as the above procedure.

The fastening of strainer plates can alternatively be carried out by tackwelding, preferably by TIG-welding.

Riveting (for stainless or mild steel backing plates):

Strainer plates can also, as by the previous procedure, be fastened by means of rivets after drilling and counter sinking of rivet holes in backing plates, see drawing 41-33432.

The rivet holes in the strainer plate should preferably be made by means of a drill, having a diameter a little smaller than that of the rivet. After drilling, countersinking takes place as indicated on drawing 41-33432. The holes might also be made by punching with a pointed drift. By this method, however, there is a risk that the strainer plate may rise around the hole during the riveting procedure.

Hammer the strainer well in order to avoid clearance between backing and strainer plate.

Surplus material of the rivet heads and the strainer plate edges must be well trimmed.

Removal of Press Screws:

Ref. main drawing.

When handling the press screws and other heavy parts of the press, it is important that the lifting tackle is sufficiently powerful. The weight of the different parts is noted on the special list provided.

When the inlet chute and upper parts of the press cage have been removed as described previously, remove the fixing screws from the flanges of the gearbox output shafts. Flanges and screw bodies have corresponding marks to facilitate later reassembling. If these marks are illegible, marks should be renewed before dismantling is carried out.

Inlet end plate and cover plate for frame are removed and the connection between bearing houses and carrier undone. Take care of seal house, end plate/supporting plate and bearing housing/carrier.

Put a sheet of gasket or similar material (approx. 3 mm thick) under the press screw, between the flights and the second strainer plate (counted from the inlet end). Slacken the hoist until the screws are resting on the gasket.

Loosen the gearbox output shafts from the press screws by means of the dismantling screws (supplied) by inserting these in the threaded holes of the shaft flanges. During this operation both forcing screws should be turned at the same rate. Take care of the dowels, which are fitted between flange and screw body.

When flange and press screws are disconnected, make sure that there is a clearance between the flange and the press screw before lifting the latter.

Place a safe cable rope sling around one of the press screws at its centre of gravity and lift it gently out of the press. Lowering should be done on a prepared place, preferably on wooden beds, avoiding damage to the screw flights.

All handling of the heavy parts should be arranged safely and with great care.

If bearings and seal house are to be dismantled from the shaft, this can be done after the press screws have been removed from the press. Particular consideration to seals and gaskets must be exercised. To ease refitting, the positions on the journal should be marked before they are detached. The position of the roller bearings should also be marked. See that the seal rings are facing the right way when refitting.

Remounting of Press Screws:

It is presumed that bearings, housing and seal ducts are mounted on the press screw journal before refitting the press screw.

The press screw is lifted at its centre of gravity and lowered into the press cage so far that only a small clearance exists between bearing housing and frame.

The screw is then axially abutted to the gearbox output shaft flanges. The flights of one screw should run midway between the flights of the other.

The flange screw bolts are pulled to 2200 Nm (1660 ft.lbf) and locked by locking washers.

The end plate and seal house are then mounted. Remember the dowels for the end plate. The bearing housings are filled with grease according to lube chart.

When checking the fitting, turn the screws by turning the gearbox input shaft by hand some turns, then afterwards by motor.

Replacement of oil seals on gearbox output shafts:

All item Nos. refer to section drawing of gearbox and drawing No. 41-33429

The "Split Klosure" oil seal (item 44) can be replaced by loosening the screws on the oil seal cover (item 14) and the screws in the supporting ring (item 16) and pulling the cover and ring including the two V-rings away from the oil seal housing.

If any of the V-rings (item 45) are to be replaced, the press screws must be disconnected from the gearbox and pushed away from the flange about 80 mm (3"). Remember first to disconnect the end plate, seal ducts, retainer and the inlet end bearing as these will have to follow the screws rearwards.

The end plate must be lifted off and the seal ducts swung outwards. The press screws are displaced by means of dismantling screws in the flanges. Make sure the inlet bearing housings are free.

The upper cage outlet end should be taken off. The cover (item 14) is detached and the defective V-rings are cut off and removed. The new V-rings can be fitted to the output shaft by stretching until they slip over the flange and the covers (items 15 and 14). Make sure the sealing lip faces the right way. The V-rings should be oiled or greased and treated carefully in order to avoid damage.

The stretching and slipping over the flange should be carried out by two men, one on each side of the flange. A piece of rag should be wrapped around the V-ring, protecting this when sliding it over the edge of the flange. The V-ring should be stretched evenly.

A pair of wooden levers can be used to lever the V-ring over the flange.

After having replaced the V-ring the confines should be filled with grease. This is done by undoing both of the plugs in the cover (item 14) and fitting a standard grease nipple in one of the holes. Then grease is pressed in by a grease-gun until it flows out of the other hole. Now, refit the plugs. It is not necessary to lubricate during normal running, but it is recommended to change grease once a year.

Dismantling of Gearbox:

All item Nos. refer to section drawing of gearbox.

When the gearbox is to be dismantled, the screw connection between press screws and gearbox output shafts has to be disconnected. The screws are displaced rearwards about 35 mm freeing the guide on the flange from the screws. To displace the screws it is necessary at the inlet end to disconnect the seal ducts, end plate, bearings and seal retainers.

The end plate for the cover is to be removed.

The pipe from the oil pump, with gauge and filter is detached. The output shaft oil seal housing (item 13) is loosened from the gearbox, and is moved somewhat outwards on the shaft.

Seal retainers and seal covers (item 13 and 14) can be left assembled for the time being.

In recesses behind the seal boxes, 6 springs (item 31) are positioned. The function of these is to locate the outer bearings when the press is running idle. The springs are now loose. Bearing covers for the screw shafts should be taken off.

Screw bolts in the flanged connection between top and bottom part of the gearbox should be undone. (2 of the bolts are fitted bolts).

On the side facing the press the bolts are internally accessible through scuttles on top of the gearbox. The top section should be lifted gently away, avoiding damage to internal pipework etc. Then the two output shafts including flanges, spur wheels, gland houses and bearings can be lifted out. The bearings should be covered up. Bearing covers for shafts No. 3 are loosened. The screws in the flange between the gearbox mid-section and bottom section are undone (2 of the bolts are fitted bolts). Further, the bolts connecting gearbox and frame are removed (6 bolts). The mid section can now be lifted away.

The two shafts No. 3, complete with bearings and gear wheels, are removed. The bearings should be covered up. The shafts must be marked, ensuring repositioning in the previous location.

The two shafts No. 2 (item 24), can be lifted out complete with gear wheels – this when bearing covers, bearings and bearing housings have been removed. The shaft should be marked. For removal of input shaft, the bearing cover, bearing and bearing house have to be removed before the shaft can be taken out. To facilitate the dismantling, all shafts have grooves for oil pressure connection.

Assembly of Gearbox:

See drawings Nos. 41-31751 and 41-33429.

All machined surfaces on gearbox must be cleaned and traces of old gasket material should be removed. Roller bearings should be well cleaned. Mating flanges between the gearbox sections should be made leakproof by means of suitable gasket compound (nonhardening type Permatex No. 2 or similar), applied in a thin even layer. See that no gasket compound material gets into the bearings and the gearbox bearing recesses.

When fitting the gearbox main sections the fitted bolts should be watched to ensure correct fitting.

When the output shafts are put in place, the teeth of gear wheels should engage in such a way that the mutual position of press screws is correct. This is when the match-marks on the flange and screw coincide and the flights of one screw are running midway between the flights of the other.

Avoid damage to the V-rings (item 45) and the Split Klosure rings (item 44) during assembly. Then seal housings, covers, support rings and seals should be taken apart and fitted individually. See drawing No. 4I-33429. Remember the springs (item 31).

The confines around the V-rings should be filled with grease. See section about replacement of seals on the gearbox output shafts.

Fitting of Roller Bearings:

Most of the bearings in the gearbox and the inlet end bearings are mounted on tapered sleeve.

When fitting roller bearings by means of tapered sleeve, it is of utmost importance that the fitting of the sleeve is done correct.

The reduction of radial clearance (clearance between rollers and the race) is an expression for how tight the bearing is fitted.

It is therefore necessary to measure the clearance before and after the fitting of the sleeve, and check that the reduction of clearance is according to tabulation given in the end of this section.

Before measuring, the bearing should be turned a few times to make sure the rollers are in position. Further the bearing should be free on the shaft without external loads when the measurement is conducted.

The radial clearance is measured by means of feeler gauge pulled hard through the bearing.

Alternatively axial displacement of sleeve can be found. However, this method is unprecise and is only recommended when accessibility prevents the radial clearance to be measured.

The tapered sleeve is driven in by considerate blows by hammer on the sleeve rim (item 19 and 65). Use a piece of hard wood between hammer and sleeve.

SKF or Stord International A/S can supply hydraulic equipment for dismantling and refitting of tapered bearing sleeves.

The following tabulation shows the required clearances and the corresponding axial displacement when fitting SKF spherical roller bearings with tapered inner race.

Metric:

Bore diameter mm	Reduction of radial clearance mm	Axial displacement taper 1/12		Min. remaining clearance mm	Bearing designation
		min mm	max mm		
80-100	0.045-0.060	0.70	0.90	0.035	22220 CK
100-120	0.050-0.070	0.75	1.10	0.050	22224 CK
120-140	0.065-0.090	1.10	1.40	0.055	23128 CK

Imperial:

Bore diameter	Reduction of radial clearance	Axial displacement taper 1/12		Min. remaining clearance	Bearing designation
		min	max		
3¼" - 4"	0,0018"-0,0025"	0,027"	0,035"	0,0014"	22220 CK
4" - 4¾"	0,0020"-0,0028"	0,030"	0,045"	0,0020"	22224 CK
4¾" - 5½"	0,0025"-0,0035"	0,045"	0,063"	0,0022"	23128 CK

Removal and refitting of shafts in the lower section of gearbox

All items Nos. refer to section drawing of gearbox.

General:

The three shafts in the lower section can be removed and replaced without having to dismantle the upper sections or shafts.

Before starting the dismantling, all bearing covers should be marked to ensure correct repositioning. Outlet chute for press cake will have to be removed to allow accessibility to the bearings on the output side of the gearbox.

Removal of input shaft (item 25):

Both of the bearing covers are removed. The washer on the rear side of the gearbox (item 19) is detached, and the tapered sleeve is pulled off by tightening up the removal nut (SKF KM 22) (delivered with the press). Bearing covers (item 9) inclusive bearing and snap ring are taken away.

The shaft inclusive of bearing (item 36) can now be pulled out. This can be done together with the bearing housing (item 8).

Removal of shaft No. 2 (item 24):

See drawing No. 41-31767.

Gearwheels (items 29 and 30) must be well supported against the bottom of the gearbox by means of wooden props.

Bearing covers, end washers, tapered sleeves, bearings and bearing houses in both ends should be detached as described for the input shaft. The stopper for the key (item 81) is removed.

The gearwheel is well propped against the gearbox wall by means of two wooden blocks and the shaft is pressed out through the gearwheel hub as illustrated by drawing No. 41-31767.

Insert and enter the extraction bolts through a thick steel plate which has a jack between this and the end of shaft (shown in principle on drawing No. 41-31767). The shaft may then be pressed out by means of the nuts or the jack. The two shafts have to be taken out on alternate sides of the gearbox. Seen gearbox face, the right shaft will have to come out through the rear side and the left through the front.

Refitting of shaft No. 2:

See drawing No. 41-31767.

The shafts are positioned in their respective openings in the gearbox (bearing housings not fitted yet). Right hand shaft (seen facing gearbox front) is taken through the rear side, the left through the front. The gearwheel is well supported against the wall in the gearbox by means of wooden blocks.

The shaft is pressed into the gearwheel hub by means of six extraction bolts, as seen on drawing No. 41-31767.

The wheel is locked on the shaft by the stopper (item 81) which is well secured.

Gearwheel and shaft are displaced axially, until the wheel is close to the wall, where it is supported by means of wooden blocks.

One of the shaft journals is now sticking out of the gearbox and is accessible for mounting of bearings. When the bearing housing with snap ring (items 9 and 43) has been fitted to the gearbox, the bearing with tapered sleeve and end washer (items 35, 37 and 19) can be fitted to the shaft journal. When fitting bearings, the instructions must be closely followed. See section about mounting of roller bearings.

The shaft inclusive the gearwheel and bearing is now guided axially inwards until the outer race of bearing trust against the snap ring in the bearing housing.

The bearing in the other end of the shaft is fitted and then the corresponding bearing housing. Finally the bearing covers (item 10) are fitted. For both shafts No. 2 the same procedure of assembly is followed.

Remember fiber gaskets for bearing covers and housings.

Repositioning of input shaft:

The input shaft (item 25) including fitted bearing (item 36) is guided into the gearbox through the opening in the front wall.

The bearing housing and bearing in the rear wall is mounted as described for shaft No. 2.

Bearing housing in the front wall is fitted.

When the bearing cover has been put on, ensure that the oil seal (item 46) is not damaged.

Stord Twin Screw Press Type RS-64SF**Data sheet****1. Weight, Volume and Principal Dimensions:**

Total weight net, approx.	20,6 tons	(23,0 US tons)
Overall length of press	7,38 m	(24,5 ft)
Max. width of press	1,97 m	(6,5 ft)
Total height of press	2,17 m	(7,1 ft)

2. Drive and Power:

The overall reduction in the gearbox is 125 : 1.

Speed Input Screws	Shaft	Consumption of power, approx.
3,5 rpm	437 rpm	Recommended
4,5 rpm	563 rpm	motor size
5,5 rpm	687 rpm	90 kW

3. Type of Motor:

The motor should have a power rating allowing ample reserves of power.

For fixed speed drive by 3 ph. squirrel cage motor, the starting torque should be at least twice the normal running torque, to enable start of fully charged press.

For variable speed drive, power allowance must be made for various losses caused by inefficient methods of speed regulation. An overall ratio of speed regulation of 1 : 3 is recommended.

4. Quantity of Lubricating Oil:

Gearbox	220 Litre	(48 Imp. gals/ 58 US gals)
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(quality according to the lubrication chart)

Stord Twin Screw Press**MS-64SF****Weight of parts**

	Approx. net weights	
Cover for cage,	400 kg	(880 lbs)
Press cage, two upper parts	3 970 kg	(8 800 lbs)
Press Screws, each	2 350 kg	(5 200 lbs)
Inlet chute	100 kg	(220 lbs)
Outlet chute	70 kg	(150 lbs)
Middle & rear supports for frame	300 kg	(660 lbs)
Gearbox, complete	6 300 kg	(13 900 lbs)
Gearbox, upper section	800 kg	(1 800 lbs)
Gearbox, middle section	950 kg	(2 100 lbs)
Gearbox, lower section	1 010 kg	(2 200 lbs)
Gearbox output shaft incl. spur wheel, flange and bearings, each	1 120 kg	(2 500 lbs)
Intermediate spur wheel incl. shaft, bearings and pinion, each	410 kg	(900 lbs)
Small spur wheel incl. shaft bearings and pinion, each	200 kg	(450 lbs)
Input shaft with bearings	75 kg	(170 lbs)
Press frame incl. cage lower part and sling clamps	8 200 kg	(18 000 lbs)
Transmission parts	140 kg	(300 lbs)

No.	Description:	Item:	Drawing/ Material:
1	Press screw, left	1	21-37841-C
1	Press screw, right	2	21-37841-C
24	Hex.head screw M 36 x 100, galvanised	3	21-37841-C
12	Locking washer	4	41-34177-B
8	Sling clamp, inlet end	5	31-35769-B
8	Bolt for sling clamp, inlet	6	41-32800-B
16	Snap ring, Seeger, A 35 x 1,5, phosphor-bronze	7	41-32800-B
26	Sling clamp, outlet	8	31-35771-C
26	Bolt for Sling clamp, outlet	9	41-33497-B
52	Snap ring, Seeger, A 50 x 2, phosphor-bronze	10	41-33497-B
34	Screw for sling clamp	11	41-33496-B
1	Bridge No. 1, lower part (outlet bridge)	12	21-42087-A
1	Bridge No. 1, upper part (outlet bridge)	13	21-42087-A
10	Bridge No. 2 - 10 and 12, lower part	14	21-42087-A
10	Bridge No. 2 - 10 and 12, upper part	15	21-42087-A
1	Bridge No. 11, lower part	16	21-42087-A
2	Bridge No. 11, upper part	17	21-42087-A
1	Bridge No. 13 at cone, lower part	18	21-42087-A
1	Bridge No. 13 at cone, upper part	19	21-42087-A
1	Bridge No. 14 at cone, lower part	20	21-42096-A
1	Bridge No. 14 at cone, upper part	21	21-42096-A
2	Bridge No. 15 and 16, lower part	22	21-42096-A
3	Bridge No. 15, 16 and 17, upper part	23	21-42096-A
1	Bridge No. 17, lower part	24	21-42096-A
1	Backing plate at end bearing	25	31-38979-B
4	Backing plate L = 934	26	11-34367-A
4	Backing plate L = 799	27	11-34367-A
4	Backing plate L = 888	28	11-34367-A
4	Backing plate L = 638	29	234279-B
2	Backing plate, L = 328, upper left - lower right fra inlet	30A	234275-B
2	Backing plate, L = 328, upper right - lower left fra inlet	30B	234275-B
4	Backing plate L = 348	31	234275-B
4	Backing plate L = 768	32	234275-B
2	Frame for strainer plate under inlet	33	21-37532-B
4	Strainer plate L = 952	34	234279-B
4	Strainer plate L = 816	35	234279-B
4	Strainer plate L = 906	36	234279-B
4	Strainer plate L = 656	37	234279-B
4	Strainer plate L = 410	38	234275-B
4	Strainer plate L = 367	39	234275-B
4	Strainer plate L = 788	40	234275-B
2	Strainer plate L = 657	41	21-37532-B
16	Distance ring	42	234279-B
10	Distance ring	43	234275-B
2	Distance ring	44	234275-B
1	Central bar, lower L = 3323	45	21-42093-A
1	Central bar, upper L = 2673	46	21-42093-A
1	Central bar, lower L = 1786	47	21-42093-A

0.8

1.2

2.5

3.0

No.	Description:	Item:	Drawing/ Material:
1	Central bar, upper L = 1200	48	21-42093-A
1	Central bar, upper L = 650	49	21-42093-A
2	Central bar, upper and lower, L = 367	50	21-42093-A
1	Side bar, lower left, L = 2673	51	21-42094-B
1	Side bar, lower left, L = 650	52	21-42094-B
1	Side bar, lower right, L = 2673	53	21-42094-B
1	Side bar, lower right, L = 650	54	21-42094-B
1	Side bar, lower left, L = 1140	55	21-42094-B
1	Side bar, lower right, L = 1140	56	21-42094-B
1	Side bar, lower left, L = 646	57	21-42094-B
1	Side bar, lower right, L = 646	58	21-42094-B
2	Side bar, lower left - upper right, L = 359	59	21-42094-B
2	Side bar, lower right - upper left, L = 359	60	21-42094-B
1	Side bar, upper left, L = 2673	61	21-42094-B
1	Side bar, upper right, L = 2673	62	21-42094-B
1	Side bar, upper right, L = 650	63	21-42094-B
1	Side bar, upper left, L = 650	64	21-42094-B
1	Side bar, upper left, L = 1200	65	21-42094-B
1	Side bar, upper right, L = 1200	66	21-42094-B
83	Socket head cap screw, M 16 x 35	67	21-42093-A/21-42094-B
41	Socket head cap screw, M 12 x 35	68	21-42093-A/21-42094-B
2	Stay L = 2700	69	11-43927-B
2	Stay L = 2180	70	11-43927-B
4	Hex. head nut M 30	71	11-43927-B
32	Distance tube	72	11-43927-B
1	Frame with press liquid tray	73	11-43918-A
4	Support under frame	74	31-34782-B
8	Hex.head screw M 30 x 120	75	31-34782-B
8	"Nylock" lock nut M 30, galv.	76	31-34782-B
2	Distance bolt	77	31-38947-A
34	Hex.head screw M 24 x 55, w/spring washer	78	11-43918-A
2	Hex.head screw M 16 x 55, w/spring washer	79	11-43918-A
1	Flange $\varnothing 280^{\circ} 160 \times 20$	80	11-43918-A
8	Hex.head screw M 20 x 65, m/mtr.	81	11-43918-A
1	Inspection glass $\varnothing 160$	82	231549-B
1	Outlet chute	83	31-31551-B
1	Gear box, 3-sectioned, i = 1 : 125	84	11-32359-B
1	Air filter, type EK 2", brass	85	21-32054-B
1	Inlet chute	86	11-43920-B
1	End plate, split	87	21-42097-A
2	Stuffing box	88	31-38977-A
1	Cover plate for frame, inlet	89	31-38976-A
4	Joint ring, "Walkersele"	90	11-43929-B
2	Bearing house SNH 524-620	91	11-43929-B
2	Spherical roller bearing, 22224 EK	92	11-43929-B
2	Split taper socket H 3124 w/nut and lock washer	93	11-43929-B
2	Double lip seal, TSNA 224G	94	11-43929-B
4	Nipple for grease gun, 1/4" thr., straight	95	11-43929-B
2	End cover ASNH 524-620	96	11-43929-B
2	Bearing sleeve	97	41-39807-B

 Stord International a.s. BERGEN		Press MS-64SF		Nr. 41-39985-A	
DATO 14 Feb. 94		SIGN. EK/LIS		BL. NR. 3	ANT. BL. 4
		List of parts for drwg. No. 11-43928		ERSTATNING FOR	ERSTATTET AV

No.	Description:	Item:	Drawing/ Material:
4	Hex.head screw M 24 x 100 w/nut	98	11-43929-B
4	Washer for M 24	99	11-43929-B
8	Hex.head screw M 12 x 55	100	11-43929-B
26	Hex.head screw M 12 x 20	101	11-43929-B
9	Hex.head screw M 10 x 15	102	11-43929-B
4	Clamping sleeve $\varnothing 6 \times 70$	103	11-43929-B
2	Clamping sleeve $\varnothing 4 \times 24$	104	11-43929-B
2	Distance ring $\varnothing 4$	105	11-43929-B
2,4 m	Rubber joint 40 x 3	106	41-30270-B
3	Cover, topp section	107	21-40028-B
1	Top section, outlet end	108	21-40038-B
2	Side hatch, outlet end	109	21-40039-B
4	Side hatch, inner	110	21-40031-B
6	Side hatch, external	111	21-40032-B
1	End plate for gear box, i = 1 : 125	112	21-40021-B
24	Mounting screw for cover, M 10 x 20	113	11-40694-B
12	Hex.head screw M 10 x 20	114	11-40694-B
24	Lock handle	115	11-40694-B
8	lock angle, compl. w/screw, nut and washer	116	31-37806-B
6	Hex.head screw M 30 x 80	117	11-32359-B
4	Hex.head screw M 36 x 120	118	11-32359-B
4	Hex.head lock nut, M 36	119	11-32359-B
1	Squirrel cage motor	120	11-32075-C
1	"Delta" pump, type JSDG, $\frac{3}{4}$ "	121	11-32075-C
1	Coupling, Eupex, type BT-80	122	11-32075-C
1	Cover guard	123	41-31462-B
4	Hex.head screw M 10 x 25, galv.	124	11-32075-C
6	Spring washer $\varnothing 10$, galv.	125	11-32075-C
2	Hex.head screw M 10 x 16, galv.	126	11-32075-C
6	Hex.head screw M 12 x 30, galv.	127	11-32075-C
8	Spring washer, $\varnothing 12$	128	11-32075-C
1	Baseplate	129	31-32237-B
1	Strainer, EPE S 75, 1"	130	11-32075-C
1	Pipe 1", L = ca. 205	131	11-32075-C
2	Back nut 1"	132	11-32075-C
2	Hex.head screw M 12 x 45	133	11-32075-C
1	Union coupling, 1"	134	11-32075-C
3	Pipe joint, 1" x $\frac{3}{4}$ "	135	11-32075-C
1	Pipe union, $\frac{3}{4}$ " inner/outer	136	11-32075-C
1	Pipe $\frac{3}{4}$ " L = ca. 1175	137	11-32075-C
1	Flanged cross, $\frac{3}{4}$ "	138	11-32075-C
1	Plug, $\frac{3}{4}$ "	139	11-32075-C
1	Thermometer 0 - 100°C, $\frac{1}{2}$ " Thr.	140	11-32075-C
1	Transition joint, $\frac{3}{4}$ " x $\frac{1}{2}$ "	141	11-32075-C
1	Pressure gauge 3F/63, 0-16 bar, R $\frac{1}{4}$ ", w/glycerine	142	11-320075-C
1	Pressure gauge cock, $\frac{1}{4}$ " inner/outer	143	11-32075-C
1	Transition joint, $\frac{1}{2}$ x $\frac{3}{4}$ "	144	11-32075-C
1	T-pipe, $\frac{3}{4}$ x $\frac{1}{2}$ x $\frac{3}{4}$ "	145	11-32075-C
2	Pipe joint, $\frac{3}{4}$ "	146	11-32075-C
1	Union coupling, $\frac{3}{4}$ "	147	11-32075-C

No.	Description:	Item:	Drawing/ Material:
1	Magnetic filter EPE, ES 125 MUE - LU, 1"	148	11-32075-C
1	Union coupling, 3/4"	149	11-32075-C
1	Pipe, 3/4" L = 245	150	11-32075-C
2	Back nut, 3/4"	151	11-32075-C
2	Plug, 1 1/4"	152	11-32075-C
	Oil head tank with piping	153	21-32054-B
	Flush water piping	154	11-43937-A
3	Hex.head screw M 20 x 65 m/ mutter	155	11-43927-B
3	Spring washer, Ø20	156	11-43927-B
4	Hex.head screw M 30 x 65	157	31-34782-B

Stord Twin Screw Press

MS-64SF

List of Tools

Item	Part		To fit	No. required
1	Wrench 46M		Sling Clamps	1
2	Ring Wrench	19 x 24	Nuts & bolts misc.	1
3	Socket Wrench	WM 39 (19)	Int. Hex. head Screws	1
4	Socket Wrench	WM 37 (17)	Int. Hex. head Screws	1
5	Socket Wrench	WM 34 (14)	Int. Hex. head Screws	1
6	Socket Wrench	WM 30 (10)	Int. Hex. head Screws	1
7	Socket Wrench	WM 28 (8)	Int. Hex. head Screws	1
8	Socket Wrench	WM 26 (6)	Int. Hex. head Screws	1
9	Socket Screw, fully threaded	M24 x 130	Extraction screws for flanged connection gearbox/press screws	2
10	Removal Nut	SKF KM 22	Roller Bearings	1
11	Hook Wrench	SKF HN 22	Roller Bearings	1
12	Hook Wrench	SKF HN 20	Roller Bearings	1
13	Removal Nut	SKF KM 30	Roller Bearings	1
14	Impact Wrench	SFK Nr. 718909	Roller Bearings	1
15	Extended Socket Wrench WM 39 (19)		Int. Hex. head bolts inside gearbox	1
16	Spanner	13 x 17	Hex. head bolts	1
17	Spanner	19 x 24	Hex. head bolts	1
18	Spanner	30 x 36	Hex. head bolts	1
19	Spanner	46 x 55	Hex. head bolts	1

Spanner Nos. 1, 10-15 are delivered with the press.

Stord Twin Screw Press MS-64SF List of Spare Parts

It is recommended to keep the following spare parts in stock:

Item No.	Description	Req. for		Ref. No.
		1-2 presses	3-4 presses	

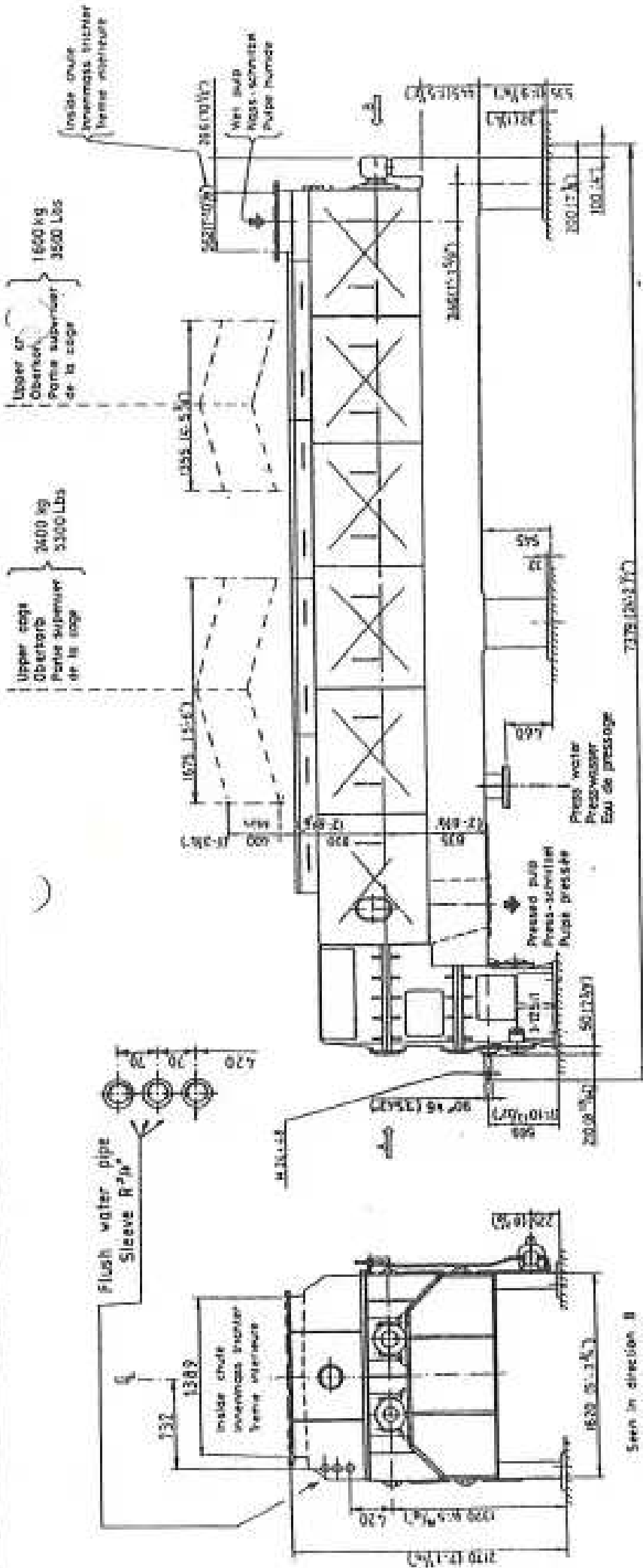
Press Section Drawing No. 11-43928-A:

26	Backing plate	L = 934	1	2	11-34367
27	Backing plate	L = 799	1	2	11-34367
28	Backing plate	L = 888	1	2	11-34367
29	Backing plate	L = 638	1	2	234279
30A	Backing plate	L = 328	1	2	234275
30B	Backing plate	L = 328	1	2	234275
31	Backing plate	L = 348	1	2	234275
32	Backing plate	L = 768	1	2	234275
34	Strainer plate	L = 952	4	8	234279
35	Strainer plate	L = 816	4	8	234279
36	Strainer plate	L = 906	4	8	234279
37	Strainer plate	L = 656	4	8	234279
38	Strainer plate	L = 410	4	8	234275
39	Strainer plate	L = 367	4	8	234275
40	Strainer plate	L = 788	4	8	234275
41	Strainer plate	L = 657	4	8	21-37532
	Rivets for strainer plates		1/2 set	1 set	
90	Seals "Walkersle", Part No. 57 - 140		2	4	11-43929
94	Double lip seals, TSNA 224G		2	4	11-43929
10	Snap ring A50 x 2 DIN 471		30	60	41-33497

Re.: Drawing No. 41-31751 Gearbox:

25	Shaft with pinion, stage 1	T18 - M6	0	1	31-31325
24	Shaft with pinion, stage 2	T16 - M8	0	2	31-31299
26	Pinion shaft, stage 3,	T18 - M13	0	2	31-31330
44	Oil seal, Split Clozure		4	8	Gaco 238, 1 x 200
45	Oil seal		2	4	V 200
46	Oil seal		2	4	Angus SM 951201:
	Filter cartridge		2	4	

Spare parts will usually be kept in stock at our workshop in Norway, or could be delivered at short notice.

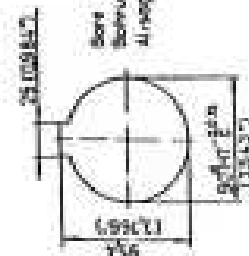


Seen in direction B
near B but quarter
from NW direction B

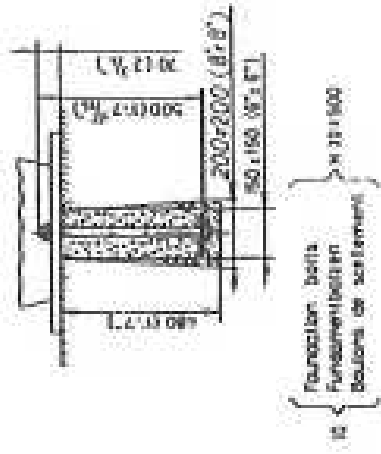
Total weight, net
Geometric mean, net
Ponds total, net

Total weight incl. pulp
Gesamtgewicht einsch. pulp
Poids total y compris les p.

	26000 kg	57350 lbs
Total weight incl. pulp		
Gesamtwegicht einsch. Pulp		
Poids total y compris les pulpes		
	Loss elimination and centre of gravity stored on foundation plate	
	Gesamtwegung und Schwerpunkt unter gleichm. Fundamentplatte	
	Separation du poids et centre de gravité sur le plan de fondation	



Sow in spring pulley
Schwung in der harmonischen
Umschlag der Pulley



Seien im directions A
Von A aus gehen
Vus im direction A

water targets are included.
Gargantuan's then wanders amongst
Dome's brides, coming

Dimensions in mm (inch)
 Height in mm and (inch)
 Dimensions mm et (pouce)

